Seminar

Performance Analysis: widening options with gem5.

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Initial configuration

The initial parameters configuration of the code was the following:

- **Number of cores:** 1
- **L1 Data & Instruction response latency:** 2 cycles
- **L2 Cache size:** 256kB
- **Matrix size:** 32

Main metric to consider: Initial **execution time:** 0.054 ms

Certain parameter variations and correspondent results

Table 1 shows the new execution times for certain modifications in the system.

Table 1: Modifications of the system parameters

Parameter	Modification	Execution time (ms)
Number of cores	2	0.192
	4	0.097
	8	0.120
L1 Caches latency (response)	20	0.092
	200	3.670
	2000	11.235
L2 Cache size	Half (32kB)	0.054
	Double (128kB)	0.054
Matrix size	Half (16)	0.010
	Double (64)	0.347
	Four times (128)	0.723

Discussion

- Number of cores

The increase on the number of cores results in a higher parallelization, therefore reducing the execution time. On the other hand, if the matrix size is not too big this run time is actually worsen since the more cores, the more competition.

- L1 (Data and Instruction) Caches Response Latency

In the end, increasing some latency in both the data and instruction caches result in an overall higher latency and obviously a larger execution time, whether it is the tag, the data or the response latency. Besides, other metrics get affected too:

    - Reduction of the overall misses
    - Increase of miss latency
    - Reduction of accesses
    - Increase of average miss latency

- L2 cache size

While a bigger L2 Cache size helps on its hit ratio and therefore access latency, In this case it does not seem to make a great difference due to the size of the matrix.

- Matrix size

The number of components directly affects the amount of computations, therefore decreasing or increasing the execution time as expected.